

Manuel Alejandro Cardoso Duarte

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25 years



Embedded Software Engineer

Bachelor of Technology

Formation

- 2017–2022 **Bachelor of Technology. Thesis: "Analysis of thermal cycles dynamics produced by light in binary mixtures for nanofluids" (presented)** , *Universidad Nacional Autónoma de México (UNAM)*, Centro de Física Aplicada y Tecnología Avanzada (CFATA).
- 6 months in an exchange program in Spain at the Universidad Autónoma de Madrid, Superior Polytechnic School (Escuela Politécnica Superior), a scholarship from the UNAM-Santander program.
 - 1 month in an exchange program to improve English skills and learn about the New England culture at the UNAM-Boston Center for Mexican Studies.
 - 2 months on a research internship in Spain at Applied Photonics Group from the University of the Basque Country - UPV/EHU, "working on the development of optical fiber sensors".
 - Winner of the Gabino Barrera medal, a distinction awarded to the highest GPA of the academic year for academic excellence.

Professional experience

- October 2022 – Today (2 years) **Embedded Software Engineer**, *KOSTAL AUTOMOTIVE SERVICES MEXICANA, SA DE CV*, Working with door control modules (DCU) and other products for the Automotive Industry.
- Develop embedded software in the C programming language for the application Software Components (SWC) and Basic Software (BSW) following AUTOSAR architecture and KOSTAL Frameworks. Understand AUTOSAR architecture to implement Software Components using Vector Developer and customize Basic Software modules using Vector Configurator.
 - Review, analyze, and implement the software requirements of the system of customers such as Stellantis, FORD, and Rivian, using platforms such as DOORS and CDB to develop efficient embedded software designs.
 - Understand, maintain, and fix problems in existing source code in the development process.
 - Configure CAN and LIN bus communications through AUTOSAR BSW COM Stack (CanDrv, CanIf, CANSM, COMM, DBC integration) to comply with customer requirements. Analyze CAN logs provided by the client when a problem is reported.
 - Working on the sleep wake-up process on the application and BSW side, understanding and configuring the AUTOSAR state machine on the shutdown process (ECUM, BSWM rules) and sleep state (low power consumption).
 - Integration and configuration of AUTOSAR memory stack (NVM, FEE, and FLS). Use the MemAnalyzer tool to analyze memory dumps during the development and post production phase.
 - Understand the Microcontroller documentation, for most of the projects the supplier is Infineon (TraveoII family) and configure the MCAL using the EB-TRESOS.
 - Follow coding standards such as MISRAC.
 - Attend an on site testing session when an issue is reported by a client.
- August 2022 – November 2022 **Research Assistant**, Center for Applied Physics and Advanced Technology (CFATA), UNAM.
- Analysis of different data of the experiments to build models using the finite element method.
 - Data analysis of the data acquired from the experiments using Matlab, Python, and C++.
- February 2018 - september 2025 (partial shift, Weekends) **Team Coach**, Engineering Faculty, Secretaría de Extensión y Cultura Universitaria, Universidad Autónoma de Querétaro (UAQ).. Working with a team of high school students to participate in the robotic competition "World Robot Olympiad" (WRO)..
- Teach students how to use the Robotic Kits and programming in languages such as Python and C to solve engineering problems and apply that knowledge at the competition.

February 2022 – **Robotics Teacher at** , "Workshop" Engineering Faculty, UAQ, Teaching high school students robotics, math, and physics.
June 2022

skills

Programming Languages C/C++, Java, Python, Assembler, $\text{\LaTeX}$, Bash, Wolfram Mathematica
IDE's Visual Studio Code, Eclipse, NetBeans, Code::Blocks, Pycharm, Jupyter, MATLAB, MATLAB Simulink, Mathematica, DaVinci
Microcontroller experience Infineon TraveoII, Texas instruments, STM32, Raspberry Pi, Nvidia Jetson
Spanish Native Speaker
English C1 Advanced
Capabilities Teamwork, Leadership
Hobbies swimming, reading

Articles

Communications Physics **Unveiling the properties of liquids via photothermal-induced diffraction patterns.**
link <https://www.nature.com/articles/s42005-023-01278-x>
Optics in 2023 **Optical Signatures of Liquids .**
link <https://opn-web-afd-d3bfk5bcc5asbs.z02.azurefd.net>

Projects

2024 – today **Robotics Coach.**, Guide a team to develop an Autonomous vehicle to participate in the annual competition World Robot Olympiad (WRO).

- Work with students to build an Autonomous Vehicle following the WRO Future Engineers Category rules.
- Generate a YOLO model to detect the semaphores on the field which the robot should use to follow the expected trajectory.
- Use the YOLO model with a Raspberry Pi 5 board and AI Hat which uses the Hailo 8 accelerator to track the object in real time.
- The progress can be seen on the GitHub repositories hyperlink: WRO Future Engineers 2026 and WRO Future Engineers 2025

2021 – 2022 **Research student at the photonics laboratory (CFATA), UNAM.**, Working in different tasks and experiments in the laboratory using photonic equipment of high performance.

- Synchronization of different equipment to obtain data over the experiments, such as oscilloscopes, spectrometers, cameras, and other devices.
- Analysis of different data from the experiments to build models, using Python libraries like OpenCV to process images.
- Instrumentation of different equipment in the laboratory.

may 2022 – July 2022 **Research internship at the Applied Photonics Group from the University of the Basque Country - UPV/EHU.**, Working in different tasks and experiments in the laboratory using photonic equipment of high performance.

- Learning about different technologies and methods for designing optical fiber sensors.
- Learning how to use interrogators to analyze samples and obtain the optical spectrum.

August 2019 – May 2021 **Research Student at UNAM, Optimization of search algorithms using architectures of high performance (GPU) in maps of different countries using C and C++ programming languages**, two projects.

- Developing algorithms based on metrics such as travel, time and energy in maps of different countries to find the shortest path between regions.
- Use GPU architecture to speed up algorithm efficiency (parallel programming).

- January 2019 – **Research student at the Center for Applied Physics and Advanced Technology (CFATA), UNAM, *System control of the 3D Bio-printer with Cartesian configuration.***, 3D-printer axis development using standard G-code for position control and python for the interactive Interface, they were presented on XX Technology Colloquium, UNAM.
- Design the electronic part using Raspberry Pi as the central computer.
 - Development of the user interface using Python.
- August 2018 – **Research student at the Institute of Neurobiology (INB), UNAM, *Cloning of sequences of subunits of the receptor GABA of rats***, The neurotransmitter GABA acts through the activation of channel-type receptors called GABA. To characterize the receptor expressed in oligodendrocytes, isolate and coding the sequence of the subunits, $\alpha 1$ was cloned, and the sequences of the subunits $\alpha 3$, $\beta 2$, and $\gamma 3$ were mutated. Presented at XIX Technology Colloquium, UNAM
- September 2017 – **Research student at CFATA, UNAM, *Porcine Bio-hydroxyapatite obtained by alkali hydrothermal, defatted, and deproteinized.***, The project consisted of standardizing the technique of obtaining bio-hydroxyapatite from mammalian bones, which involves removal of the organic phase through processes of degreasing using the Soxhlet method and deproteinized in an alkaline solution. The removal of the organic phase was registered by infrared spectroscopy. Presented at XVIII Technology Colloquium, UNAM

Achievements and awards

- November 7th, 2023 **Coach of one team on the final phase of the Robotics Competition "World Robot Olympiad", Panama city.**
- Preparation of the team to compete in the Robotics Competition, Robomission category.
- September 10th, 2022 **Coach of one team in the National Robotics Competition "World Robot Olympiad", Mexico city.**
- Preparation of the team to compete in the Robotics Competition.
 - Third Place in the category "RoboMission".
- September 10th, 2022 **Coach of one team in the National Robotics Competition "World Robot Olympiad", Mexico city.**
- Preparation of the team to compete in the Robotics Competition.
 - Third Place in the category "RoboMission".
- June 27th, 2022 **The ICFO International School on the Frontiers of Light: New horizons in Quantum Materials, Spain, ICFO - The Institute of Photonic Sciences.**
- student at the one-week school.
- August 9th, 2019 **International Robotics Competition "Robotchallenge", Beijing, China, University of Science and Technology Beijing.**
- First Prize, Free Style, the project presented "Implementation of an embedded system for a proprioceptive Virtual Reality interface".
 - Second place, "Silver Medal" in the category "Line follower".
- September 17th, 2017 **National Robotics Competition "World Robot Olympiad", Monterrey, México, Monterrey Institute of Technology and Higher Education.**
- Participation in the competition.
- August 4th, 2017 **International Robotics Competition "Robotchallenge", Beijing, China, Zhongguancun National Independent Innovation Demonstration Zone Exhibition Center.**
- First place in the category "Lego Unknown Mission-Senior".
 - Second place in the category "Line Follower LEGO-Senior".
 - Third place in the category "Lego Sumo-Senior".
- January 26th, 2017 **National Olympiad Biology Competition, Mexican Academy of Sciences, Campeche, México.**
- Third Place, "Bronze medal", of the competition.
- November 2016 **International Robotic Competition Robomatrix, Uniminuto institute, Bogota, Colombia.**
- First Place Line Follower LEGO.
 - First Place Megasumo Lego.